MazG (Q88MB7 / PP_1657) in *Pseudomonas putida* KT2440: A House-Cleaning Nucleoside Triphosphate Pyrophosphohydrolase

Summary

MazG (UniProt Q88MB7, locus PP_1657) of *Pseudomonas putida* KT2440 is a cytoplasmic, divalent-metal-dependent nucleoside triphosphate pyrophosphohydrolase (NTP-PPase; EC 3.6.1.8) that catalyzes the hydrolysis of nucleoside triphosphates to their corresponding nucleoside monophosphates plus inorganic pyrophosphate (NTP + H_2O → NMP + PP_i). It belongs to the MazG-like all-α NTP-PPase superfamily (Pfam PF03819; InterPro IPR011551), a group of "house-cleaning" enzymes whose principal physiological role is to sanitize the cellular nucleotide pool by removing non-canonical and oxidatively damaged (deoxy)nucleoside triphosphates before they can be misincorporated into DNA or RNA. Characterized MazG orthologs hydrolyze all eight canonical ribo- and deoxyribonucleoside triphosphates, with a preference for deoxynucleotides, and additionally destroy mutagenic species such as 8-oxo-dGTP, dUTP, and 2-hydroxy-(iso-)ATP.

No primary experimental study has been published on the *P. putida* KT2440 protein itself. However, the functional assignment for Q88MB7 rests on an unusually strong chain of homology-based evidence. Global sequence alignment shows Q88MB7 is **56.7 % identical (74.7 % similar) to the biochemically and structurally characterized *Escherichia coli* MazG** (P0AEY3), whose crystal structure and NTPase mechanism were solved by Lee et al. (2008). Q88MB7 is a 277-residue, **two-domain (tandem) enzyme** with duplicated MazG cores (residues 28–101 and 180–236), and it retains the conserved catalytic **E-x-GDLLF** metal-coordinating motifs in both lobes. An AlphaFold model confirms this tandem architecture at high confidence (overall mean pLDDT 88.5), with both α-helical catalytic cores modeled at very high confidence and a lower-confidence flexible inter-domain linker.

At the genomic level, PP_1657 lies immediately downstream of *relA* (PP_1656, the (p)ppGpp synthetase), preserving a *relA*–*mazG* synteny seen across many bacteria and pointing to a role in tuning the **stringent-response alarmone ppGpp** and nucleotide homeostasis. Notably, and unlike *E. coli*, the *P. putida* locus **lacks an adjacent *mazEF* toxin–antitoxin module**, so the *E. coli*-specific link between MazG and *mazEF*-mediated programmed cell death should not be assumed for this organism. In short, MazG in *P. putida* is best described as a soluble cytoplasmic nucleotide-pool-sanitizing enzyme coupled, through genomic and biochemical context, to the stringent stress response.

Gene / Protein Identity Verification

Before presenting findings, the identity of the target was verified against the UniProt record, satisfying the mandatory checks in the research brief:

Attribute	Expected (UniProt Q88MB7)	Verified in this investigation
Gene symbol	*mazG*	□ Consistent with MazG-family assignment
Organism	*Pseudomonas putida* KT2440	□ Locus PP_1657 confirmed in KT2440 genome
EC number	3.6.1.8	□ Reaction NTP → NMP + PP_i confirmed for family
Protein family	MazG-like NTP pyrophosphohydrolase	□ PF03819 / IPR011551, duplicated cores
Key domains	MazG-like N + C (IPR048015, IPR048011)	□ Two tandem MazG domains (res 28–101, 180–236)

Conclusion of verification: The gene symbol *mazG* is *not* ambiguous in this case. The literature for MazG-family NTP pyrophosphohydrolases aligns precisely with the UniProt description, domain architecture, and EC number for Q88MB7. The one caveat is that *mazG* is genomically associated with the *mazEF* toxin–antitoxin system in *E. coli*, and some literature conflates MazG with programmed cell death; this investigation explicitly checked and found that the *mazEF* module is **absent** adjacent to PP_1657, so that association does not transfer to *P. putida*. Research therefore proceeded on the correct gene.

Key Findings

1. MazG is a nucleoside triphosphate pyrophosphohydrolase (EC 3.6.1.8) that hydrolyzes NTP → NMP + PP_i

The primary, defining function of MazG is the hydrolysis of a nucleoside triphosphate into a nucleoside monophosphate plus inorganic pyrophosphate. This reaction (EC 3.6.1.8) is established by direct biochemical characterization of multiple MazG orthologs. *E. coli* MazG was shown to hydrolyze **all eight of the canonical ribo- and deoxyribonucleoside triphosphates** to their respective monophosphates and PP_i, with a notable **preference for deoxynucleotides** (PMID: 12218018). The *Thermotoga maritima* ortholog (Tm-MazG) similarly "catalyzes the hydrolysis of all eight canonical ribo- and deoxyribonucleoside triphosphates to their corresponding nucleoside monophosphates and PP_i," and in a second step **hydrolyzes the resultant PP_i to P_i**, giving it an additional pyrophosphatase activity (PMID: 12657645).

Q88MB7 is assigned to this family and reaction by UniProt (EC 3.6.1.8; MazG-like NTP-PPase superfamily; PF03819 / IPR011551). Because the reaction chemistry and broad substrate range are conserved features of the enzyme family and are supported by the very high sequence identity to characterized enzymes (see Finding 7), the *P. putida* protein is confidently assigned the same catalytic activity.

Reaction:

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 NTP   +  H2O   ──MazG (Mg2+/Mn2+)──►   NMP   +  PPi
(dNTP)                                 (dNMP)
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2. MazG is a "house-cleaning" enzyme that removes non-canonical / oxidatively damaged (d)NTPs

Beyond turning over canonical nucleotides, the biologically important role of MazG is to act as a **"house-cleaning" enzyme** — one that eliminates aberrant nucleotides from the pool so they cannot be incorporated into nucleic acids, where they would be mutagenic or cytotoxic. A structure-guided classification placed dimeric dUTPases, HisE, and MazG in a new superfamily of all-α NTP pyrophosphohydrolases and predicted that "uncharacterized members of this superfamily perform 'house-cleaning' functions by hydrolyzing abnormal NTPs and are functionally analogous to the structurally unrelated hydrolases of the Nudix superfamily"

(PMID: 15740738). The same analysis pointed to **2-hydroxy-adenosine (isoguanosine) triphosphate — a product of oxidative damage of ATP — as the most likely substrate** for the *Sulfolobus* MazG active site.

This prediction was confirmed experimentally in mycobacteria: mycobacterial MazG "is a potent NTP-PPase capable of hydrolyzing all canonical (d)NTPs, as well as the mutagenic **dUTP** and **8-oxo-7,8-dihydro-2'-dGTP**" (PMID: 20529853). Because 8-oxo-dGTP and 2-hydroxy-ATP arise from reactive oxygen species and are strongly mutagenic if incorporated into DNA, MazG's ability to hydrolyze them ties the enzyme directly to genome-protection and oxidative-stress defense. This house-cleaning role is the most likely primary physiological function of *P. putida* MazG.

3. MazG adopts a metal-dependent, all-α-helical fold with a conserved acidic active site

Crystal structures of MazG orthologs show a conserved all-α architecture built from a characteristic four-helix module. The *Bacillus*-conserved MazG "assembles into a tetrameric architecture. Each monomer adopts a four-α-helix bundle that accommodates a metal ion using four acidic residues, and presents one putative substrate-binding site" (PMID: 26920050). This confirms two hallmarks of the family: (i) an all-α fold and (ii) a **metal-coordinating cluster of acidic residues** (glutamates/aspartates) at the active site that positions the catalytic divalent cation (Mg^{2+} or Mn^{2+}).

Crucially for Q88MB7, the *E. coli* enzyme — the closest characterized homolog — is a **two-domain monomer**: "composed of two similarly folded globular domains in tandem. Among the two putative catalytic domains, only the C-terminal domain has well ordered active sites and exhibits an NTPase activity" (PMID: 18353782). This tandem two-domain organization matches the domain annotation for Q88MB7 exactly (see Finding 5) and indicates that in the *P. putida* enzyme, catalysis is likely concentrated in the ordered C-terminal MazG core, with the N-terminal core playing a structural/regulatory or accessory role.

4. MazG regulates nucleotide/alarmone (ppGpp) homeostasis and supports survival under nutritional and oxidative stress

MazG's enzymatic activity has a documented downstream consequence for the **stringent response**. In *E. coli*, "enzymatic activity of MazG in vivo affects the cellular level of guanosine 3',5'-bispyrophosphate (ppGpp), synthesized by RelA under amino acid starvation. The reduction of ppGpp, caused by MazG, may extend the period of cell survival under nutritional stress" (PMID: 18353782). MazG thereby acts as a counterweight to RelA — trimming the ppGpp alarmone and modulating how long cells persist during starvation.

MazG also physically engages the translation/ribosome-associated GTPase machinery: "we discovered that Era interacts with MazG" — a direct protein–protein interaction with the essential GTPase Era (PMID: 12218018). In *E. coli*, *mazG* is "being transcribed in the same polycistronic mRNA with *mazEF*" (PMID: 16390452), placing it in the operon context of the *mazEF* toxin–antitoxin/programmed-cell-death module in that organism. Finally, the stress relevance of MazG is functionally demonstrated in mycobacteria, where "deletion of *mazG* in *M. smegmatis* rendered the mycobacteria defective in response to oxidative stress" (PMID: 20529853). As a soluble metabolic enzyme acting on cytosolic nucleotide pools, MazG carries out its function in the **cytoplasm**.

5. Q88MB7 is a 277-aa two-domain enzyme with duplicated PF03819 cores and conserved E-x-GDLLF metal-binding motifs in both lobes

Domain analysis of UniProt Q88MB7 shows a **277-residue protein containing two MazG-like NTP-pyrophosphohydrolase domains** (residues 28–101 and 180–236). Pfam PF03819, Gene3D 1.10.287.1080, and SUPFAM SSF101386 are each annotated in **duplicate** ("2"), and the protein maps to eggNOG **COG3956** (MazG family). Sequence analysis identifies the conserved MazG metal-coordinating acidic signature **E-x-GD-x-x-F in both domains** — E59-LGDLLF in the N-terminal lobe and E203-VGDLLF in the C-terminal lobe — each flanked by additional acidic residues (…ERGDF…, …DGDADALED…) that form the metal-binding cluster.

This tandem two-domain organization mirrors *E. coli* MazG (a two-domain monomer in which catalytic activity resides in the C-terminal domain; PMID: 18353782) rather than the single-domain minimal MazG proteins such as *T. maritima* TM0360 (PMID: 25758186) or *D. radiodurans* DR2231 (PMID: 21733847). This is an important classification: it means Q88MB7's closest functional model is the well-studied *E. coli* enzyme, whose broad canonical-NTP substrate range and ppGpp-modulating physiology are the most directly transferable.

6. In KT2440, *mazG* (PP_1657) lies immediately downstream of *relA* (PP_1656) but lacks an adjacent *mazEF* module

Genomic-neighborhood analysis (KEGG) shows PP_1657 (*mazG*) is directly adjacent to **PP_1656**, annotated as ATP:GTP 3'-pyrophosphotransferase, KEGG orthology **K00951 = GTP pyrophosphokinase / (p)ppGpp synthetase RelA** (EC 2.7.6.5; UniProt Q88MB8), within the purine-metabolism pathway. The other immediate neighbors — PP_1654 (cysteine synthase), PP_1655 (RlmD rRNA methyltransferase), and PP_1658–1660 (conserved proteins of unknown function) — **do not include *mazE* or *mazF* toxin–antitoxin genes**.

This is a significant organism-specific distinction. In *E. coli* the chromosomal arrangement is *relA–mazEF–mazG*, embedding *mazG* in the programmed-cell-death operon. In *P. putida* the ***relA–mazG* adjacency is retained but the intervening *mazEF* module is absent.** The conserved *relA–mazG* linkage strengthens the interpretation that MazG participates in tuning the ppGpp stringent response (consistent with Finding 4), while the missing *mazEF* module means the *E. coli*-specific programmed-cell-death connection should **not** be extrapolated to *P. putida*.

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E. coli:      relA ── mazE ── mazF ── mazG      (mazG in mazEF operon)
P. putida:    relA ── mazG                       (relA-mazG synteny; no mazEF)
           (PP_1656)  (PP_1657)
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7. Q88MB7 is a clear ortholog of the biochemically and structurally characterized *E. coli* MazG (56.7 % identity, 74.7 % similarity)

A global Needleman–Wunsch alignment (BLOSUM62) of Q88MB7 (277 aa) against *E. coli* MazG P0AEY3 (263 aa) yields **148/261 identical positions (56.7 % identity) and 74.7 % similarity across essentially the full length**. P0AEY3 is the enzyme whose crystal structure and NTPase mechanism were solved by Lee et al. (2008) — the "MazG-ATP complex structure and subsequent mutagenesis studies explain the peculiar active site environment accommodating all eight canonical NTPs as substrates" (PMID: 18353782).

Both tandem MazG domains are conserved between the two proteins. The N-terminal metal-binding motif **GDLLFQVV is identical**, and the C-terminal catalytic-domain signature aligns residue-for-residue:

Region	*E. coli* MazG (P0AEY3)	*P. putida* MazG (Q88MB7)
N-terminal metal motif	GDLLFQVV	GDLLFQVV (identical)
C-terminal catalytic signature	LEEE-MGDLLFATVNLARHL	LEDE-VGDLLFAAVNLARHL
Downstream MazG signature	ANxKFERRFR	ANxKFERRFR (conserved)

At **56.7 % identity**, well above the ~30–40 % threshold generally accepted for confident transfer of enzyme function and identical reaction chemistry, and with all catalytically essential acidic residues conserved, Q88MB7 can be assigned the same substrate range and mechanism as *E. coli* MazG with high confidence.

8. AlphaFold confidently predicts the tandem two-domain MazG fold

The AlphaFold DB model for Q88MB7 (AF-Q88MB7-F1, v6, 277 residues) has a **high overall mean pLDDT of 88.5 (median 91.9)**; 93.1 % of residues are modeled at confident level (pLDDT > 70) and 61.7 % at very high confidence (> 90). The two MazG-like domains are each predicted with very high confidence — **N-terminal domain (res 28–101) mean pLDDT 93.8** and **C-terminal domain (res 180–236) mean pLDDT 93.9**. In contrast, the **inter-domain segment (res 102–179) is lower** (mean 79.6, min 50), consistent with a flexible linker, and the extreme C-terminal tail (237–277) is partly disordered (min pLDDT 33).

This structural prediction independently corroborates the two-domain, all-α architecture inferred from sequence, and mirrors the experimentally observed *E. coli* two-domain fold. The lower-confidence linker is expected for a flexible connector between two rigid catalytic lobes.

Mechanistic Model / Interpretation

Integrating the eight findings yields a coherent model for the role of MazG in *P. putida* KT2440:

Molecular function. MazG is a Mg^{2+}/Mn^{2+}-dependent NTP pyrophosphohydrolase that cleaves the α–β phosphoanhydride bond of nucleoside triphosphates, releasing PP_i and the nucleoside monophosphate. It is broad-specificity toward all eight canonical (d)NTPs (deoxynucleotide-preferring) but, physiologically, its most important substrates are aberrant nucleotides.

Cellular role — nucleotide-pool sanitation. The enzyme's central job is "house-cleaning": destroying non-canonical / oxidatively damaged nucleotides (8-oxo-dGTP, dUTP, 2-hydroxy-ATP/ isoguanosine triphosphate) before DNA/RNA polymerases can incorporate them. By converting these to monophosphates, MazG removes their triphosphate "currency," preventing mutagenic misincorporation and preserving genome integrity — functionally parallel to the Nudix hydrolases (e.g., MutT) but structurally unrelated.

Regulatory coupling — the stringent response. MazG's genomic position immediately downstream of *relA* and the demonstrated ability of MazG activity to lower ppGpp levels connect nucleotide catabolism to the stringent response. RelA synthesizes the alarmone ppGpp during nutritional/oxidative stress; MazG activity trims ppGpp, modulating the depth and duration of stringent shutdown and, in *E. coli*, extending survival under starvation. The retention of *relA–mazG* synteny in *P. putida* suggests this regulatory partnership is conserved.

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Oxidative / nutritional stress
         │
         ▼
ROS ──► damaged (d)NTPs (8-oxo-dGTP, 2-OH-ATP, dUTP)
         │                                   ┌────────────────┐
         ▼                                   │  RelA (PP_1656) │──► ppGpp ↑
     ┌─────────┐                             └────────────────┘         │
     │  MazG   │ hydrolyzes damaged (d)NTPs        (genomically          │
     │(PP_1657)│──► NMP + PPi                         adjacent)          ▼
     └─────────┘         │                        MazG NTPase ──► ppGpp ↓
         │               ▼                        (tunes stringent response,
         ▼      clean nucleotide pool              extends survival)
  genome protection      │
(prevents mutagenic      ▼
 misincorporation)   faithful DNA/RNA synthesis
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Localization. As a soluble enzyme acting on cytosolic nucleotide pools, MazG operates in the **cytoplasm**. There is no signal peptide or transmembrane region implied by its all-α globular fold.

What differs in *P. putida* vs. *E. coli*. The *E. coli* literature ties MazG to *mazEF*-mediated programmed cell death because of operon co-transcription. In *P. putida*, the *mazEF* module is absent adjacent to PP_1657. Therefore, the *P. putida* enzyme is best interpreted as a nucleotide-sanitizing / stringent-response-modulating enzyme, and claims about programmed cell death should be treated as *E. coli*-specific and not automatically transferred.

Evidence Base

PMID	Study (organism / system)	How it supports the findings
12218018	*E. coli* MazG; Era interaction	Defines EC 3.6.1.8 reaction; all 8 canonical (d)NTPs, deoxynucleotide preference; MazG–Era interaction (F1, F4)
12657645	*T. maritima* Tm-MazG	Confirms NTP→NMP+PP_i for all 8 NTPs; extra pyrophosphatase activity (F1)
15740738	Superfamily classification	Predicts house-cleaning role; 2-OH-ATP as substrate; all-α superfamily (F2)
20529853	Mycobacterial MazG	Experimental hydrolysis of dUTP & 8-oxo-dGTP; oxidative-stress defect on deletion (F2, F4)
26920050	*Bacillus* MazG crystal structure	Tetrameric all-α fold; metal-coordinating acidic active site (F3)
18353782	*E. coli* MazG crystal structure	Two-domain tandem architecture; catalytic C-terminal domain; ppGpp modulation; reference for homology transfer (F3, F4, F7)
16390452	*E. coli* mazEF/mazG regulation	*mazG* co-transcribed with *mazEF* in *E. coli* (F4; contrast with F6)
25758186	*T. maritima* TM0360 structure	Single-domain minimal MazG; contrast to two-domain Q88MB7 (F5)
21733847	*D. radiodurans* DR2231	dUTP-specific MazG-like house-cleaning enzyme; oxidative-stress framework (F2, F5)
36937295	Housecleaning MazG structural analysis	Reinforces house-cleaning nomenclature/role for the family (F2)
26075750	Human DCTPP1 (MazG-like)	Illustrates substrate-specialization within superfamily; tetrameric form (context)
21972224	*E. coli* cisplatin stress proteomics	MazG among stress-responsive proteins (context)
33265965	Oxidative-stress transcriptomics	Broader oxidative-stress-response context (context)

Concordance and tension. The core reaction and house-cleaning role are supported by concordant biochemistry across at least three organisms (*E. coli*, *T. maritima*, mycobacteria) plus structure-based prediction. The main point of tension is the *mazEF*/programmed-cell-death association, which is documented in *E. coli* (PMID: 16390452) but does **not** apply to *P. putida* because the *mazEF* module is genomically absent there — an important negative finding that prevents over-interpretation.

Supported and Refuted Hypotheses

Supported (homology + strong cross-species evidence): - **H1** — Q88MB7 is an NTP pyrophosphohydrolase (EC 3.6.1.8) catalyzing NTP → NMP + PP_i. *Supported.* - **H2** — It acts broadly on canonical (d)NTPs with deoxynucleotide preference. *Supported.* - **H3** — It performs house-cleaning removal of damaged nucleotides (8-oxo-dGTP, dUTP, 2-OH-ATP). *Supported.* - **H4** — It is a metal-dependent, all-α-helical enzyme with a two-lobed MazG core. *Supported.* - **H5** — It is cytoplasmic. *Supported (inference from soluble metabolic-enzyme role).*

Refined / revised for *P. putida* specifically: - **H6a** — PP_1657 is embedded in a *mazEF* toxin–antitoxin operon (as in *E. coli*). *Refuted by genomic context* — no *mazE/mazF* genes neighbor PP_1657. - **H6b** — PP_1657 is functionally linked to ppGpp/stringent-response metabolism. *Strengthened* — PP_1657 is directly downstream of *relA* (PP_1656, (p)ppGpp synthetase), conserving *E. coli relA–mazG* synteny. - **H6c** — PP_1657 participates in *mazEF*-mediated programmed cell death. *Unsupported for this strain* — the linked toxin–antitoxin module is absent.

Limitations and Knowledge Gaps

1. **No primary study of the *P. putida* protein.** Every functional statement about Q88MB7 is an inference from homology, domain architecture, genomic context, and structural prediction. No enzymatic assay, knockout phenotype, or structure exists for the KT2440 enzyme specifically.
2. **Substrate specificity is inferred, not measured.** While 56.7 % identity to *E. coli* MazG strongly implies the same broad canonical-(d)NTP range with deoxynucleotide preference, the exact kinetic parameters (kcat, Km) and the relative efficiency toward damaged

nucleotides in *P. putida* are unknown. MazG-like enzymes can specialize (e.g., DR2231 is dUTP-specific; human DCTPP1 prefers dCTP), so specialization in Q88MB7 cannot be fully excluded.

3. **Oligomeric state uncertain.** Characterized MazGs range from dimers (TM0360) to tetramers (*Bacillus*, mycobacterial) to two-domain monomers (*E. coli*). Q88MB7's two-domain architecture predicts an *E. coli*-like monomeric two-domain arrangement, but its actual quaternary structure has not been determined.
4. **ppGpp/stringent-response coupling is by analogy.** The *relA–mazG* synteny is suggestive, but a direct effect of *P. putida* MazG on ppGpp levels has not been measured.
5. **Metal preference and active-site details** (Mg^{2+} vs. Mn^{2+}; which acidic residues are catalytically essential) are inferred from motif conservation rather than mutagenesis in this ortholog.

Proposed Follow-up Experiments / Actions

1. **Recombinant enzymology.** Express and purify His-tagged Q88MB7; measure pyrophosphohydrolase activity against all eight canonical (d)NTPs plus the mutagenic substrates 8-oxo-dGTP, dUTP, and 2-hydroxy-ATP. Determine kcat/Km to confirm the predicted broad specificity and deoxynucleotide preference, and to test for specialization.
2. **Metal-dependence and active-site mutagenesis.** Test Mg^{2+} vs. Mn^{2+} dependence and mutate the predicted metal-binding glutamates (E59, E203 in the GDLLF motifs) to confirm catalytic essentiality, distinguishing the roles of the N- and C-terminal cores.
3. **Genetic knockout phenotyping.** Construct a PP_1657 deletion in KT2440 and assay sensitivity to oxidative stress (H_2O_2, paraquat), mutation frequency (rifampicin-resistance assays), and survival during amino-acid starvation, to test the house-cleaning and stringent-response roles in vivo.
4. **ppGpp measurement.** Quantify intracellular ppGpp in wild-type vs. Δ*mazG* under starvation to test whether MazG modulates the alarmone as in *E. coli*.
5. **Structure determination.** Solve the crystal or cryo-EM structure (or experimentally validate the AlphaFold model) to confirm the tandem two-domain fold, oligomeric state, and active-site geometry.

6. **Interaction mapping.** Test whether *P. putida* MazG interacts with the Era GTPase homolog and/or RelA, probing conservation of the regulatory network beyond the missing *mazEF* module.

Evidence Confidence Summary

Claim	Evidence type	Confidence
NTP pyrophosphohydrolase, EC 3.6.1.8 (NTP→NMP+PP_i)	Family/EC + ortholog biochemistry + 56.7% id to *E. coli* MazG	**High**
Broad canonical (d)NTP substrates, dNTP preference	Ortholog biochemistry (*E. coli*, *T. maritima*)	High (moderate for exact ranking)
House-cleaning of damaged NTPs (8-oxo-dGTP, dUTP, 2-OH-ATP)	Ortholog biochemistry + superfamily prediction	Moderate–High
Two-domain, metal-dependent all-α fold	Q88MB7 domains + motifs + AlphaFold (pLDDT 88.5)	**High**
Cytoplasmic localization	Inference (soluble metabolic enzyme)	High
Functional link to ppGpp/stringent response	*E. coli* data + conserved *relA*–*mazG* synteny	Moderate
NOT in a *mazEF* operon / not in *mazEF*-PCD	*P. putida* genomic context	High (for this strain)

Conclusion

MazG (Q88MB7 / PP_1657) of *Pseudomonas putida* KT2440 is, by strong homology and structural inference, a **cytoplasmic, divalent-metal-dependent nucleoside triphosphate pyrophosphohydrolase (EC 3.6.1.8)** that hydrolyzes NTPs to NMP + PP_i, acting on all eight canonical (deoxy)ribonucleoside triphosphates with a deoxynucleotide preference. Its principal biological role is **"house-cleaning" sanitation of the nucleotide pool** — destroying non-canonical and oxidatively damaged nucleotides such as 8-oxo-dGTP, dUTP, and 2-hydroxy-ATP to prevent their mutagenic incorporation into DNA and RNA — and its conserved genomic linkage downstream of *relA* implicates it in tuning the ppGpp stringent-response alarmone. Unlike *E. coli*, it is not embedded in a *mazEF* toxin–antitoxin operon, so it should not be assumed to drive *mazEF*-mediated programmed cell death. All conclusions derive from robust protein, genomic, and structural bioinformatics (notably 56.7 % identity to the mechanistically characterized *E. coli* MazG and a high-confidence AlphaFold two-domain model), as no primary study of the *P. putida* ortholog itself yet exists.
